



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Nanyang Business School

WEEK 1

Introduction to Robotic Process Automation

A manual accounts payable process

Consider which of these steps require human judgement.

09:00 Open the shared inbox and download 40 supplier invoices as PDFs

09:20 Key the invoice number, date, amount and tax into the ERP

11:00 Open the purchasing system and match each invoice to its purchase order

12:30 Copy each mismatch into a spreadsheet for follow-up

14:00 Email the approvers and follow up on outstanding responses

16:00 Update the status tracker and file the PDFs

PART ONE

What RPA is

Definitions and common misconceptions.



Definition

Robotic Process Automation is software that operates other software in the same way a person would, by reading screens, clicking, typing and copying, in order to carry out **rule-based, repetitive digital tasks** without human intervention.

Robotic

The robot is a software program running on a PC or a server. It has no physical form.

Process

A sequence of steps from trigger to outcome, rather than a single action.

Automation

The task runs on a schedule or a trigger, without a person carrying it out.

Common misconceptions

Four things RPA is not.

Not a physical robot

RPA is software. A fully automated back office contains only ordinary computers.

Not artificial intelligence

Standard RPA follows rules written by a developer. It does not learn or infer. AI can be added, but is not part of RPA itself.

Not a system replacement

RPA operates on top of existing systems. The ERP, email and websites underneath are not modified.

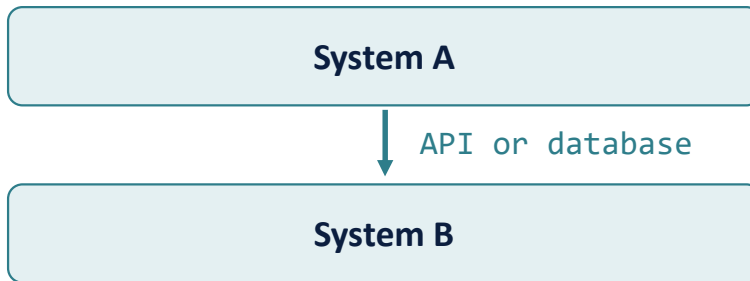
Not a remedy for a poor process

Automating an inefficient process produces the same inefficiency at higher speed. Redesign the process first.

How RPA differs from system integration

System integration

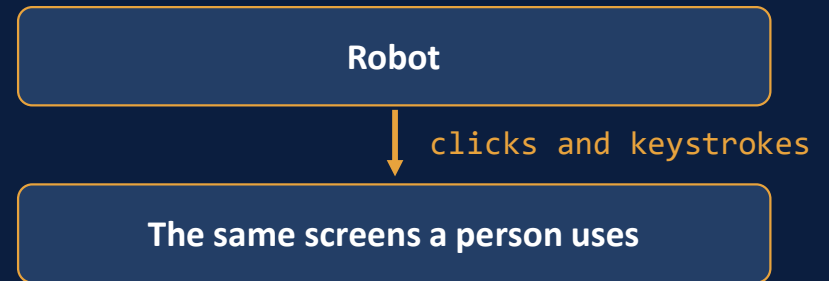
One system connects to another



Fast and reliable, but an interface must already exist and developer effort is required to build it.

Robotic Process Automation

The robot uses the user interface



Works with any application a person can operate

Comparison with other approaches

Approach	How it connects	Effort to build	Main weakness
Manual work	A person at a keyboard	None	Slow and error-prone
Excel macro	Within one application	Low	Cannot leave that application
System integration	System to system, by code	High	An interface must exist
RPA	Through the user interface	Medium	Fails when screens change
Replace the system	Rebuild the process	Very high	Cost, time and disruption

RPA requires less effort than replacing a system and has wider reach than a macro, but is less reliable than an interface.

Attended and unattended robots

Attended

- Runs on the employee's own desktop
- Started by the person, on demand
- Works alongside the person and returns the task
- Used in call centres and front office work
- Normally licensed per user

Unattended

- Runs on a server or virtual machine
- Started by a schedule, a queue or an event
- No person is present while it runs
- Used for overnight and back office processing
- Normally licensed per robot

PART TWO

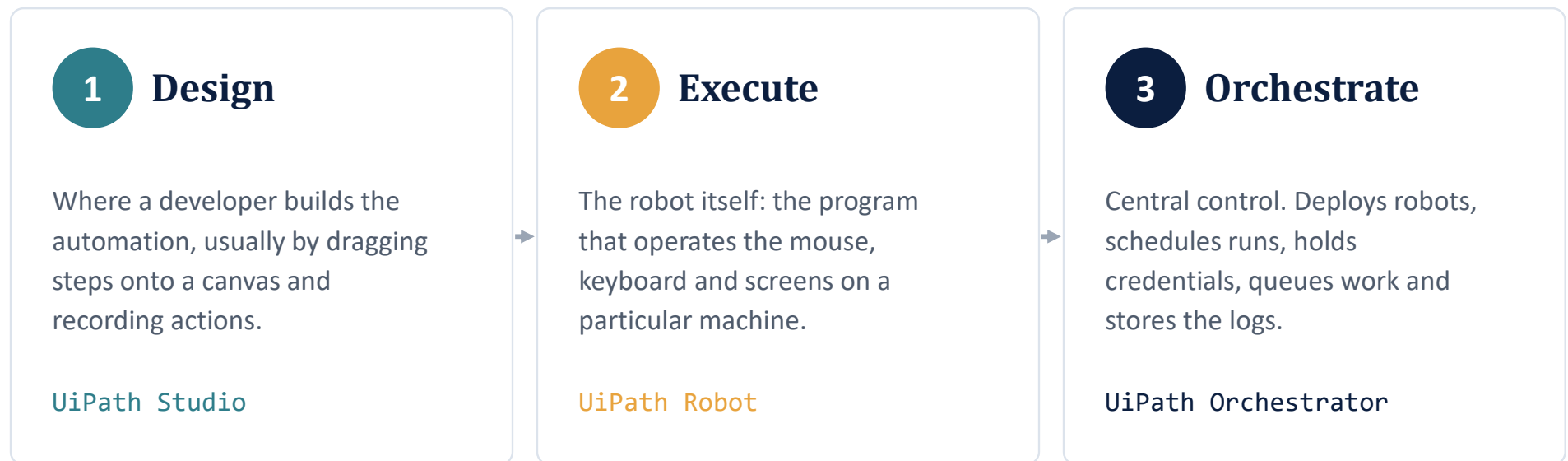
How a robot works

Platform components, selectors and the delivery lifecycle.



Components of an RPA platform

Vendors use different names for the same three components.



Orchestration provides the scheduling, credential storage and logging required for enterprise deployment.

Components of an automation

Trigger

What starts it: a schedule, a new email, a file placed in a folder, or a button.

Activities

The ordered steps, such as open browser, click, type, read cell and send mail.

Selectors

How the robot identifies a button or field on a screen.

Data

Variables and arguments that carry values between steps and workflows.

Exception handling

What the robot does when a screen does not appear or a value is missing.

Logging

A record of every step, outcome and error.

Exception handling and logging are often left out during development. Both are required before an automation is used in production.

Selectors

How a robot identifies an element on a screen.

What the user sees



Submit

What the robot uses

```
<button  
  id='btnSubmit'  
  class='primary'  
  name='Submit' />
```

A selector identifies one element on a screen. If that element changes, the selector no longer matches and the automation stops.

The vendor updates the page and renames the button

A dialog box appears that was not present during development

The window opens at a different size and the layout shifts

The screen resolution or language setting is different

The delivery lifecycle



Step 7 is often underestimated. An automation is not a project that finishes. It requires ongoing maintenance and must be budgeted accordingly.

Example: supplier invoice processing

The process shown earlier, with an automation applied.

BEFORE

- Download invoices from the inbox
- Key each invoice into the ERP
- Match each invoice to its purchase order
- Log mismatches to a spreadsheet
- Email approvers and follow up
- Update tracker and file documents

AFTER

- Robot collects and saves the files
- Robot enters the data
- Robot performs the match
- Robot produces the exception list
- Robot sends the approval emails
- Person reviews the exceptions**

PART THREE

Choosing a process

Criteria for deciding which processes to automate.

Criteria for a suitable process

High volume
+ Runs many times a day, week or month, so the build cost is recovered

Rule-based
+ Every decision can be stated as an if-then rule with no professional judgement

Structured input
+ Data arrives in a consistent digital form such as a system field or fixed template

Stable process
+ The steps have not changed recently and are not expected to change

Stable systems
+ The applications are not being upgraded or replaced in the near term

Low exception rate
+ Most cases follow the main path and exceptions can be routed to a person

Few processes meet all six criteria. Three or four is generally sufficient to proceed.

Criteria for an unsuitable process

Requires judgement

If an experienced person would answer "it depends", a rule-based robot cannot do it

Changes frequently

Rework each month will consume the saving the automation produces

Paper or free text

Scanned letters and unstructured email require document AI, not standard RPA

Low volume

Twenty transactions a year will not repay a four-week build

Unstable systems

A system upgrade will stop the automation working

Already being replaced

If a system change is funded, do not automate the existing workaround

Common applications

Finance and accounting

- Invoice processing and three-way matching
- Bank and intercompany reconciliations
- Recurring journal entries at period close
- Following up overdue receivables

Procurement

- Raising purchase orders from approved requests
- Vendor master data creation and updates
- Goods receipt posting
- Contract expiry monitoring

Human resources

- New joiner setup across multiple systems
- Payroll input validation
- Leave and claims processing
- Exit checklists and access removal

Customer and IT operations

- Order entry from emailed forms
- KYC and sanctions screening checks
- Insurance claims first-pass review
- User provisioning and password resets

Most of these processes sit between two systems that were never designed to exchange data.

[Home](#) / ... / [News](#) / Joint Operation Between The Anti-Scam Centre And Five Partnering Banks Led To Disruption Of More Than 500 Ongoing Scams

Joint Operation Between The Anti-Scam Centre And Five Partnering Banks Led To Disruption Of More Than 500 Ongoing Scams

Police News Releases | 3 Min Read | 12 May 2026

The Singapore Police Force's Anti-Scam Centre (ASC), together with five banks - DBS, UOB, OCBC, SCB and GXS, carried out a two-month joint operation that utilised **Robotic Process Automation (RPA)** for upstream detection of potential scam victims. RPA technology streamlines the exchange of information between the Police and the banks, enabling the Police to swiftly reach out to potential scam victims through SMSes.

Between 1 March 2026 and 30 April 2026, ASC officers and the participating banks sent over 5,600 SMS alerts to bank customers identified as potential scam victims, urging them to stop any further monetary transfers. The detection of these potential scam victims resulted in the successful disruption of over 500 ongoing scams and **averted potential financial losses of more than \$37 million.**

Source: https://www.police.gov.sg/media-hub/news/2026/05/20260512_joint_operation_between_the_anti-scam_centre_and_five_partnering_banks

Continuous Improvement through Process Innovation

Driven by a commitment to excellence, SGH's approach to process improvement marries innovative technology with compassionate care to provide the optimal patient outcomes and experiences.

Multifaceted Process Enhancement

Our approach to process improvement encompasses three critical areas:



Robotic Process Automation (RPA)

RPA taps on software robots to automate repetitive digital tasks and workflows. Through our implementation of RPA, SGH has achieved:

- A remarkable 40% reduction in administrative processing times
- An additional 10,000 hours annually for our dedicated staff to devote to direct patient care

This technological advancement has not only streamlined our operations but also allow us to serve our patients more efficiently than ever before.



Process Transformation & Improvement (PTI)

Our PTI team combines Lean methodologies, Design Thinking, and Change Management to drive multidisciplinary projects that optimise hospital processes. We continuously refine our workflows and system to foster a culture of improvement, ensuring we deliver the "Best Outcome, Best Experience" for both our patients and staff, reinforcing SGH's position as a trusted leader in healthcare.



Artificial Intelligence & Automation (AIA)

AIA leverages smart technology to learn and automate decision making. SGH's AIA initiatives represent our commitment to harnessing cutting-edge technology for enhanced decision-making, improved diagnostic accuracy, and personalized patient care strategies. By leveraging AI and automation, we continue to push the boundaries of modern healthcare, combining our deep sense of responsibility with an unwavering dedication to innovation.

DISCUSSION

Work in pairs

1 Identify one repetitive task you have done, at work, on internship or as a student.

2 Score it against the six criteria. How many does it meet?

3 If it were automated, what would still require a person? Be specific.

The third question is the most important of the three.

PART FOUR

Benefits, risks and control

What automation provides, what it costs, and what controls it requires.

Benefits

Speed

The robot works at machine speed and does not stop for breaks or holidays

Consistency

It applies the same rule the same way each time

Capacity

Peaks at period end are absorbed without hiring temporary staff

Traceability

Every action is logged with a timestamp

No system change

The ERP is not modified, so deployment takes weeks rather than years

Redeployment

Staff move from data entry to reviewing exceptions

The complete log of every action is of particular relevance to audit and internal control.

Limitations



Fragility

Automations stop working when screens change. A routine software update can halt several at once.



Maintenance cost

Each automation needs an owner, a test cycle and a budget. A large number requires a permanent maintenance function.



No judgement

Anything requiring interpretation or negotiation remains with a person.



Inefficiency retained

Automating an inefficient process keeps the inefficiency and makes later change harder.



Understated costs

Licences, infrastructure, developer time and support are frequently larger than the original estimate.



Staff resistance

Staff who expect to be replaced are less willing to help document the process.

Control implications

An automation accesses systems and requires the same controls as a user.

Identity

The automation needs its own named account, not a borrowed human login, or accountability is lost.

Least privilege

Grant only the access the task requires. Broad access rights create a standing risk.

Segregation of duties

An automation that both raises and approves a transaction recreates the conflict controls exist to prevent.

Change control

Automation code requires versioning, review, testing and an approved release process.

Monitoring

Failures can go unnoticed. Someone must monitor the queue and the exception log.

Continuity

If the automation stops, a manual fallback must exist and must have been tested.

The controls that apply to a human user apply equally to an automation.

Why automation programmes fail



Wrong process selected

Chosen because it was visible or convenient, not because it met the criteria



No process owner

Nobody in the business is accountable once IT has handed the automation over



No governance

Automations spread across departments with no standards, register or oversight



Maintenance unbudgeted

The build was funded but the following years of upkeep were not



Benefits overstated

Headcount savings are promised and then cannot be demonstrated



Treated as an IT project

Built without the staff who do the work, so it models the process incorrectly

Most of these are organisational rather than technical problems.

PART FIVE

The wider field

The market, the main platforms and how the field is developing.



MARKET SIZE

The scale of the RPA market

US\$35bn

global RPA market size in 2026,
from about US\$28bn a year earlier

24%

projected compound annual growth
rate to the mid-2030s

One third

of RPA spending comes from
banking, financial services and
insurance

About 60% of Fortune 500 companies use automation from a single vendor.

Sources: Precedence Research; Fortune Business Insights; UiPath FY2025 disclosures (2026 estimates).

Main vendors

UiPath

Market leader with approximately one third of global share. The platform used in the practical sessions.

Automation Anywhere

Cloud-based architecture, used for large-scale unattended deployment.

Microsoft Power Automate

Included with Microsoft 365, so often the default where that suite is already in use.

SS&C Blue Prism

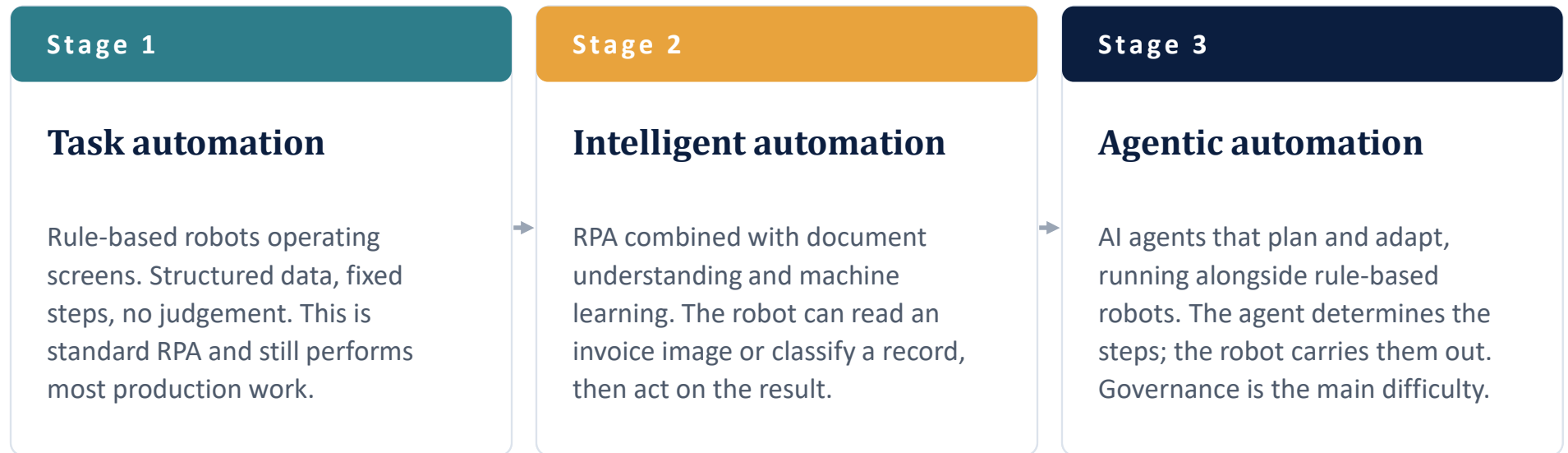
One of the earliest vendors, with a governance-focused reputation in regulated industries.

Embedded automation

SAP Build Process Automation, ServiceNow and others include automation within the platform itself.

The concepts are common across platforms. Learning one makes the others straightforward to pick up.

How the field is developing



Stage 3 does not replace Stage 1. Agents perform decision-making; robots perform repeatable execution. Current systems combine the two.

Relevant roles and skills

Roles

- Automation analyst or process analyst
- RPA developer
- Automation centre of excellence lead
- Internal auditor covering automated controls
- Finance transformation consultant

Skills

- Reading a process and identifying waste in it
- Writing rules precisely enough for a machine
- Judging what should and should not be automated
- Applying control concepts: access, approval, evidence
- Explaining a technical option to a business audience

The skills listed on the right are not specific to any one platform.

SUMMARY

Key points

- 1 RPA is software that operates other software through the user interface rather than an API.
- 2 That approach gives wide reach and quick deployment, but is less reliable than an interface.
- 3 Selecting the right process matters more than building it well. Most failures are selection failures.
- 4 An automation is a user of the organisation's systems and requires the same controls as a person.
- 5 AI has not made RPA obsolete. Agents make decisions; robots carry out repeatable steps.

Preparation for next week

Please complete this before the next session.

Bring a dataset that you need to clean, manipulate or audit.

1 If you do not have one

Ask your supervisor or a colleague for a dataset they work with.

2 Preferred format

One file with multiple tabs, rather than several separate files.

3 Ask permission

Confirm with the data owner before uploading anything to Google Drive.

4 If the data is sensitive

Anonymise it first. Remove or mask names, identification numbers and any other identifying fields.